

Test-beam topology and response-model studies of scintillator range staves for HIBEAM/NNBAR

HIBEAM/NNBAR CCB test-beam working group
Working publication package

12 August 2026

Working publication – not submission-authorising. This PDF is the canonical chapterised build surface for the paper, but the Cycle-3 adversarial audit invalidates promotion of the current ΔE – E production numbers, the July optical calibration grid, and the 8.9% deposited-energy-resolution headline. Those quantities are kept only in clearly marked publication-hold text until their physics/provenance issues close. The hold is selective: results outside those quarantine scopes, such as the component-safe B4–B6 timing pair-residual analysis and the regenerated optical-response calibration grid (a declared MC-model-dependent diagnostic), stand as final results, and each figure carries its own evidence-status note. The authoritative status is tracked in the collaboration’s claim ledger and audit records.

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Abstract

The HIBEAM/NNBAR programme is developing an annihilation detector for neutron-conversion searches, including free neutron–antineutron oscillation. One proposed detector element is a range stack of plastic-scintillator staves intended to sample the penetration depth and longitudinal energy-loss pattern of low-energy charged particles. This paper studies the CCB test-beam range-stack prototype using beam data and Geant4-based detector simulations.

The analysed B-stack readout uses channels labelled B2, B4, B6 and B8, while the corresponding detector model contains eight physical longitudinal layers. Existing source-bound selected-data products indicate different longitudinal occupancy patterns for the declared Sample-I and Sample-II run families, with one population concentrated closer to the stack entrance and the other extending more frequently to downstream readout positions. The depth profile has been regenerated from the full pre-threshold 8-channel-by-16-sample event product under the measured channel-polarity map (33 runs, 1.1M events, run-block bootstrap uncertainty): Sample I concentrates moderately at the entrance (35.3% of normalized amplitude in B2) while Sample II extends further downstream (B2 15.3%; the B8/B2 amplitude-share ratio rises from 0.71 in Sample I to 1.94 in Sample II, and the direction holds under both duplicate-readout hypotheses). An earlier 87.4%/72.7% entrance-concentration version was produced under a falsified polarity convention and is withdrawn. The quantitative result remains gated (the profile shape is threshold-dependent and the threshold/baseline contract is not yet closed), and primary run-log and hardware-trigger records are incomplete, so the difference is interpreted as a run-family observation rather than a measured trigger efficiency.

The physically relevant amplitude telescope observable is $\Delta E = A(B2)$ and $E = A(B4) + A(B6) + A(B8)$. Sparse longitudinal readout can hide the layer containing the largest energy loss and therefore distort a stopping/range proxy. The current numerical ΔE – E data/MC bundle is under publication hold because the beam event construction is threshold-censored upstream and the MC producer does not preserve an unambiguous physical-layer namespace or validated event-measure contract.

The located beam waveform product contains eight channels with 16 samples per channel. A component-safe B4–B6 analysis under the measured polarity map (Sample II, 228,697 complete pairs above the amplitude cut) gives a pair-residual central width of 0.146 ns (bootstrap interval 0.144–0.148 ns) with a -9.7 ns channel-skew median offset, superseding an 8.7 ns value produced under the falsified polarity convention. This is reported only as a pair residual at fixed amplitude selection: the constant inter-channel offset, finite 16-sample waveform format, unvalidated real-pulse component identity and timing-reference/covariance terms do not support conversion to an intrinsic single-stave resolution.

A detailed single-stave optical model provides a framework for separating quenching, scintillation, wavelength shifting, optical transport, SiPM response and electronics. The historical July optical grid and its derived held-out energy-reconstruction result are superseded model diagnostics because the campaign predates source-level optical repairs and because its target variable was Birks-visible energy rather than unquenched deposited energy. The calibration grid has since been regenerated on the repaired source with explicit E_{raw} and E_{vis} semantics, and the held-out reconstruction has been rebuilt on it with propagated uncertainty; both remain MC-model-dependent diagnostics, quarantined from validated promotion until the optical/SiPM nuisance envelope is propagated, held-out coverage spans more operating points and impact-position transfer is tested beyond the central line. Absolute beam-data-calibrated light yield and detector energy resolution therefore remain withheld.

1 Introduction

The HIBEAM/NNBAR programme at the European Spallation Source is developing the experimental techniques required for searches for baryon-number-violating neutron-conversion processes, including free neutron–antineutron oscillation [1, 3]. In a neutron–antineutron search, an antineutron would annihilate in a thin target and produce a low-energy hadronic final state whose charged particles, pions and secondary nuclear interactions must be reconstructed with high efficiency while backgrounds are rejected.

These requirements differ from those of a conventional collider spectrometer. The neutron flight region must remain magnetically quiet, restricting the use of magnetic momentum analysis close to the interaction region, while many charged annihilation products are sufficiently slow that range and specific-ionisation information are directly useful. The HIBEAM/NNBAR calorimeter concept therefore includes multiple layers of plastic scintillator ahead of the electromagnetic calorimeter, so that the longitudinal pattern of deposited energy and the depth reached by a charged particle can supplement calorimetric and timing information [2]. A segmented scintillator range stack also provides a controlled system in which the consequences of sparse readout, thresholding and finite waveform sampling can be studied explicitly.

The CCB test-beam data provide an experimental test of this range-stack concept with charged-particle events that can also be represented in Geant4. The present study focuses on five connected questions:

- (i) what longitudinal-response differences are observed between the declared beam-data run families;
- (ii) how much range and energy-loss information is lost when the analysed readout samples only alternating positions in the B stack;
- (iii) what timing observable is supported by the source-bound waveform format and estimator;
- (iv) which causal response stages must be represented in a one-fibre, one-end scintillator/WLS/SiPM model; and
- (v) how raw deposited energy and Birks-visible energy should be kept distinct when defining and validating an energy-reconstruction estimator.

The analysis intentionally separates direct beam-data observables from truth-level simulation and from model-dependent optical/electronics predictions. This distinction is essential for the present detector state: a result that is reproducible within a specified Monte-Carlo model is evidence about that model, but it becomes a detector-performance measurement only after the relevant hardware, calibration, data-lineage and held-out validation conditions are established. The evidence and provenance rules used to enforce this separation are described in the reproducibility section rather than being treated as additional detector observables.

2 CCB test-beam configuration

Hardware facts, simulation-configuration values and unresolved legacy narratives are separated in the publication hardware bill of materials. Each quantity is classified by provenance: measured, set by the design specification, set by the simulation configuration, or of unresolved external origin. Simulation agreement is not hardware provenance.

2.1 Beam, target and two-arm layout

The reviewed CCB simulation configuration uses a 190 MeV incident proton beam and a deuterated target. The model contains the elastic channel

$$p + d \rightarrow p + d,$$

and transports the outgoing charged particles toward two scintillator arms. Historical source configuration uses a nominal geometry-distance parameter of 109 cm, with the B stack near -38° and the A stack near $+71.5^\circ$, and contains eight B bars and four A bars. These values are publication-safe as simulation/configuration quantities; they are not survey-grade metrology until matched to primary collaboration records. An external collaboration preprint independently reports the in-beam CCB configuration: a 190 MeV proton beam on a CD_2 target, a TPC arm at 71° to the beam and a second scintillator arm at 38° , with two-arm coincidences selecting candidate elastic-scattering events [9]; the reported arm-angle magnitudes are consistent with the two-arm configuration above.

The compact detector simulation contains the target, scintillator stack volumes, trigger-related volumes and ProtoTPC components. Source review shows that it is useful for first-order acceptance and truth-energy-deposit studies but does not yet represent every passive material relevant to precision stopping predictions. Wrapping, supports, photosensor boards, cabling, dead layers and survey uncertainties must therefore be treated explicitly when the paper makes material-budget claims.

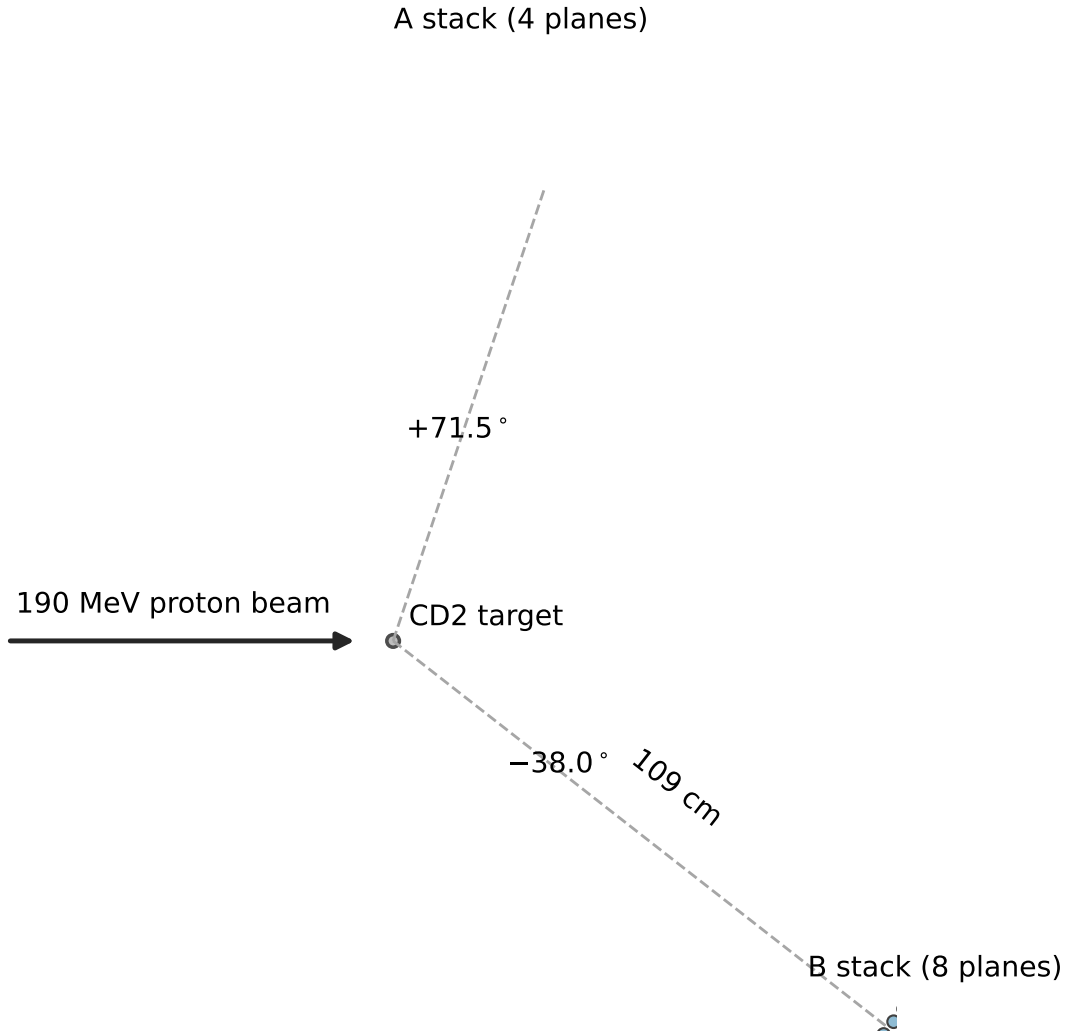


Figure 1: CCB two-arm geometry schematic. Beam energy, target, arm angles and bar counts are annotated on-figure from the source-bound simulation geometry; they are simulation/configuration values, not survey-grade metrology.

2.2 B-stack readout and run families

The analysed data use four B-stack channels labelled B2, B4, B6 and B8. The simulation contains eight physical B layers, so the data represent alternating longitudinal samples. The exact offset between detector labels and Geant4 copy/layer IDs remains a publication blocker: a historical readout-map convention argued for layers 1/3/5/7 while later design material uses 0/2/4/6. Only the every-other-layer structure is currently robust. A final data-matched MC figure must import a single source-bound detector map rather than hard-code either convention.

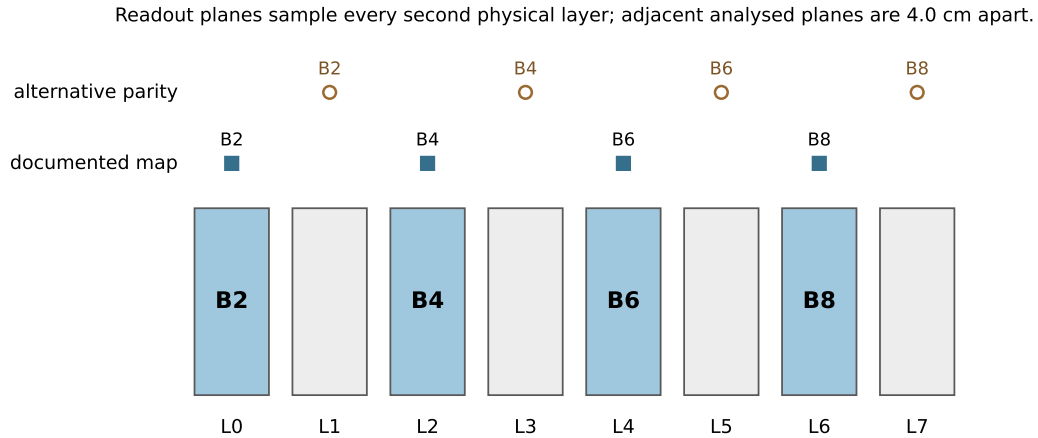


Figure 2: B-stack channel-to-layer correspondence. Channels B2/B4/B6/B8 sample every other physical Geant4 layer; the absolute offset between detector labels and copy IDs (the even-versus-odd parity) is annotated as unresolved rather than silently chosen.

The current analysis configuration separates Sample-I and Sample-II run families and also separates calibration and analysis roles. However, the exact hardware trigger thresholds, coincidence logic, prescales and run-purpose record remain partly external. The safest beam-data language is therefore *Sample-I/Sample-II run-family comparison*. The MC selection based on first-stack-layer charged hits is retained only as a Monte Carlo trigger-proxy selection until the hardware response is modelled or demonstrated equivalent.

PUBLICATION HOLD. Do not use the words “trigger efficiency”, “hardware-trigger reproduction”, or quantitative trigger-induced species fractions until the run-role record and the hardware trigger response are closed on the same run/hardware truth surface (the hardware bill of materials retains its unknown-provenance rows).

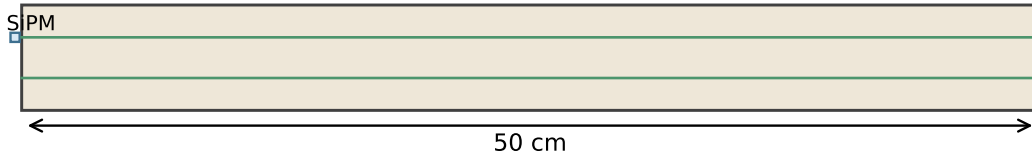
3 Scintillator stave and readout

3.1 Stave geometry

The current design specification follows a later collaboration clarification. Each B stave is described as an extruded-polystyrene scintillator approximately 50 cm long, 5.18 cm wide and 2.0 cm thick along the nominal particle path. Two longitudinal holes of 2.0 mm diameter are separated by 2.0 cm centre-to-centre and accept 1.8 mm Kuraray Y-11 wavelength-shifting fibres. The exterior includes a reflective coating. An older specification describing approximately one-metre BC-408 bars is treated as superseded external history rather than an alternative current specification.

The stave design supports two longitudinal fibres and, in the full concept, readout from both fibre ends, corresponding to four possible optical measurements. The later collaboration clarification states that the CCB beam-test configuration used one fibre at one end; in the present evidence hierarchy this remains a design-specification statement until the installed-hardware record is source-bound. If confirmed for the analysed hardware, one-ended readout makes the response dependent on longitudinal hit position and removes the charge/time asymmetry that a two-ended system could use to constrain position.

(a) longitudinal view



(b) transverse cross-section (to scale)

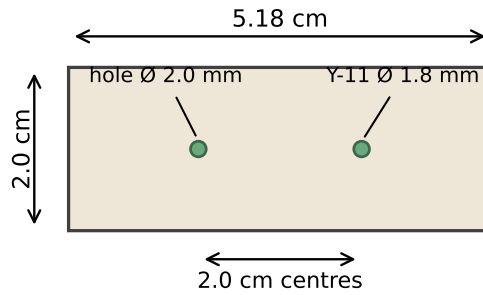


Figure 3: B-stack stave geometry (design specification). Dimensions, hole and fibre parameters and the one-fibre one-end readout configuration are annotated on-figure; of the four possible fibre-end measurements only one was instrumented, and the reflective outer coating enters the optical model as a distinct surface.

3.2 SiPM and electronics boundary

The detector model uses a Hamamatsu S13360-3050CS MPPC. Manufacturer data specify a $3\text{ mm} \times 3\text{ mm}$ photosensitive area, $50\text{ }\mu\text{m}$ pixel pitch and 3600 pixels, with spectral sensitivity covering the Y-11 green-emission region [8]. These public specifications do not by themselves determine the CCB operating response. PDE, overvoltage, temperature, optical coupling, illumination footprint, microcell recovery, crosstalk, afterpulsing, gain and the analogue impulse response are separate response objects.

The publication model therefore keeps sensor photons, primary avalanches, correlated avalanches, charge and ADC waveform observables distinct. A scalar efficiency is not allowed to absorb unresolved physics from multiple stages.

3.3 Waveform products

Two waveform schemas appear in the project history. Historical reconstruction configurations use 18 samples per channel, whereas the located LUNARC beam product contains eight channels with 16 samples per channel (128 waveform words per event). A provenance analysis classifies these as distinct schemas rather than a proven reversible 16-to-18 sample transformation. Consequently, historical sub-nanosecond timing studies based on the 18-sample product cannot be promoted to the source-bound 8-by-16 beam sample.

Every publication amplitude/timing figure must record waveform schema, raw-file hashes, producer revision, channel map and pulse-polarity contract. The publication source package intentionally does not hide these provenance requirements behind a generic label such as “raw data”.

4 Simulation programmes

4.1 CCB event simulation

The CCB event simulation provides particle identities, trajectories and energy deposits across the two-arm detector. Historical production used Geant4 electromagnetic and hadronic/ion transport suitable as a starting point for proton/deuteron studies in this energy domain [5, 6]. The simulation is used for truth-level stopping and for testing how detector segmentation changes observables.

The elastic p-d angular source table used by the current source-model work is bound to the 190-MeV centre-of-mass differential cross sections reported in Table VI of Ermisch et al. [4]. A deterministic inverse-CDF implementation defect in the source sampler has been repaired: the current source explicitly defines its interpolation and measured-support policy and can generate the nominal central-value law with unit event weights for direct-sampled campaigns. This bounded source-sampler closure must not be confused with publication-MC closure. The source-model uncertainty is now quantified at source level: a $\pm 3\%$ per-node envelope mapped to the point-to-point systematic stated by the measurement, closed against 5,000,000 common-random-number draws with the row-statistical (diagonal) replica propagation validated against the delta-method reference, and propagated to lab-frame angles by exact two-body kinematics (mean $\theta_{\text{lab}} = 39.35^\circ$, envelope band $38.81\text{--}39.89^\circ$). Production campaigns nevertheless generate central values only; campaign-level and detector-response propagation, surviving support/model alternatives, exact production provenance and downstream detector-response dependencies remain separate requirements.

Legacy samples can contain non-unit primary-event weights from a generator source-measure implementation whose frame/Jacobian semantics were later found to be defective. The legacy one-million-event paper MC file is byte-hashed but is not yet bound to a complete production receipt containing generator revision, cross-section table/mode, seed, physics list, build/toolchain and all geometry/material configuration. That legacy file therefore remains a historical provenance-gated diagnostic rather than the nominal publication MC. A regenerated 100,000-event campaign now carries a complete production receipt (generator source-pair and cross-section-table digests with the direct-CDF sampling mode, executable and runtime-stack identity, serial single-worker configuration) in a machine-readable manifest; engine seeding is recorded honestly as default per-run seeding.

4.2 Single-stave optical simulation

The single-stave model transports the charged particle through a detailed scintillator/fibre geometry and follows optical photons through scintillation, wavelength shifting, attenuation and sensor boundaries. A separate SiPM/electronics layer converts sensor arrivals into avalanche/charge/waveform quantities.

The model is scientifically most useful when represented as a causal graph:

$$E_{\text{raw}} \rightarrow E_{\text{vis}} \rightarrow N_{\gamma, \text{scint}} \rightarrow N_{\gamma, \text{WLS abs}} \rightarrow N_{\gamma, \text{WLS emit}} \rightarrow N_{\gamma, \text{sensor}} \rightarrow N_{\text{primary aval}} \rightarrow Q \rightarrow A_{\text{ADC}}.$$

Here E_{raw} is unquenched deposited energy and E_{vis} is the Birks-visible energy. These two variables must never share the same bare E_{dep} label.

The current single-stave source explicitly separates the fibre inner and outer cladding materials and records the WLS secondary-multiplicity parameter in the optical configuration. These implementation repairs post-date the July optical calibration grid used by the historical response studies. Therefore the old grid is a superseded model diagnostic; it has been regenerated from the current source (2026-08) with explicit E_{raw} and E_{vis} semantics, and promotion of absolute optical-response or energy-reconstruction numbers remains gated on propagating the optical/SiPM nuisance envelope and on broader held-out coverage.

4.3 Simulation evidence classes

Paper figures distinguish truth-level Monte Carlo, digitised Monte Carlo, model-dependent optical Monte Carlo, and calibrated-detector evidence. The last category is intentionally empty until response nuisances and held-out data closure satisfy the publication gate.

5 Data taking and event selections

5.1 Run families

Current configuration files define Sample-I and Sample-II analysis populations and separate calibration populations. These configuration roles are useful for reproducible processing but are not a substitute for the primary hardware run log. A run-role discrepancy means that calibration/analysis assignment must be bound to the exact fitted object before timing or response calibration can be described as held out.

The historical S00 selected-pulse product contains 640,737 B-stack pulses. That count is deterministic for the fixed historical producer/input set, but the claim ledger keeps this claim gated (promotion blocked) because the raw waveform width, raw-to-sorted lineage and channel-polarity/hardware contracts are not all independently closed. Selected-pulse counts are not detector efficiencies.

5.2 Longitudinal topology in the existing selected products

Existing source-bound selected-data studies show different B-stack occupancy patterns for the declared Sample-I and Sample-II populations: one population is more concentrated near the entrance while the other contains a larger downstream fraction. This is the central beam-data range-stack result. The pre-threshold 8-channel-by-16-sample event product has been regenerated and analysed: 33 runs and 1,096,728 events (Sample I 798,694; Sample II 298,034) with frozen amplitude/polarity definitions and a run-block bootstrap (1,000 replicates, seed 1318). Under the measured polarity map v2 (superseding the v1 convention falsified for channels 2–7), the normalized amplitude profiles show Sample I moderately concentrated at the entrance (B2 35.3%, B8 25.2%) while Sample II extends further downstream (B2 15.3%, B8 29.6%): the B8/B2 amplitude-share ratio rises from 0.71 (Sample I) to 1.94 (Sample II). Each stave is read out by a duplicate channel pair, so the stave assignment is carried as a nuisance envelope: under the odd-channel hypothesis the ratio rises from 0.64 to 1.28, preserving the direction. The earlier v1-based profile (87.4%/72.7% entrance concentration; $7.6\times$ deepest-stave ratio) is withdrawn as a polarity artifact. A threshold scan at 500/750/1000 ADC preserves the direction of the difference while compressing its magnitude (B8/B2 ratio contrast from $2.7\times$ at threshold 0 to $\approx 1.2\times$), so the quantitative profile remains gated in the claim ledger until a threshold/baseline contract closes.

Gated figure – not rendered in the publication PDF

Regenerated product delivered: run-block bootstrap with 1,000 replicates, a 0–1000 ADC threshold scan and a duplicate-channel (even/odd) nuisance envelope; the claim remains gated pending threshold-contract closure, so the figure is not rendered in the publication build.

Figure 4: Beam-data B-stack depth profile for Sample-I and Sample-II from the pre-threshold 8-by-16 event product under measured polarity v2 (gated).

Evidence boundary. Until the final raw-event rerun and hardware run/trigger record close, the paper should call the existing difference a run-family-dependent longitudinal-topology indication. Interpreting it uniquely as a trigger effect would confound trigger changes with possible run-period beam/electronics changes, and calling the deepest active readout a stopping layer would additionally assume a range convention that the sparse readout does not itself establish.

5.3 Monte Carlo population comparison

The corresponding MC population is formed with a Monte Carlo trigger-proxy selection. Under the corrected estimator contract (unique truth-TrackID counting on the regenerated 2M-event campaign, event-level bootstrap 16–84% intervals, provenance-bound to its input campaign by hash), the coincidence-like population is deuteron-enriched in the first B layer by track flux (59.3%, CI 57.7–60.7%) and deuteron-dominated by first-layer deposited-energy share (85.5%), while the entering-any-layer population is proton-dominated by track flux (73.4%, CI 73.0–73.7%; deuterons still carry 42.0% of its first-layer deposited energy). Track-flux, event-presence and deposited-energy compositions differ by up to 26 percentage points for the same population and are therefore reported as separate observables; a single species fraction without its estimator is not well defined. The A-arm composition is proton-led in both populations ($p \approx 65\%$) and carries negligible discriminating power between them. The proxy can show that the transport model contains physically plausible stopping/penetrating populations, but it cannot authorise a hardware-trigger efficiency or a beam-data species mixture until the trigger response is measured; the source-level cross-section uncertainty contract is now closed with a $\pm 3\%$ source-model envelope, but the campaign itself was generated at central values, so campaign-level propagation remains bounded only at source level.

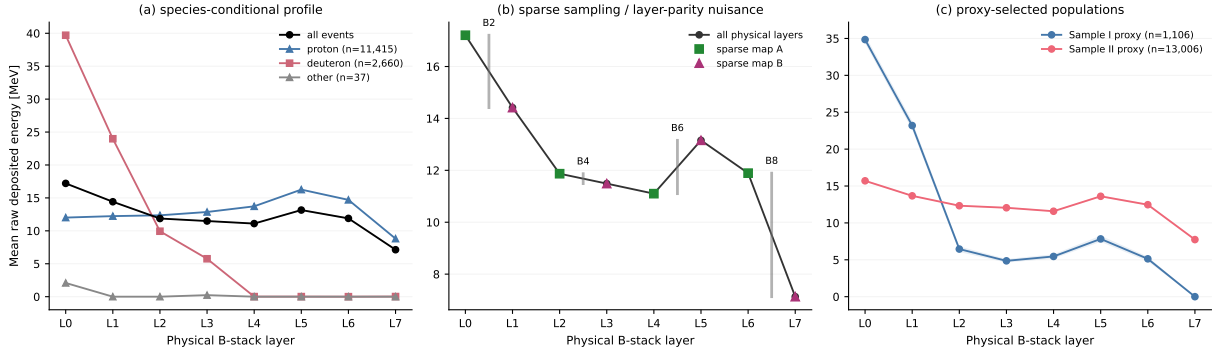


Figure 5: MC longitudinal deposited-energy profile of the proxy-selected population ($n = 14,112$, weights verified unit, $ESS = 14,112$): (a) per-layer mean raw deposited energy, species-resolved, with event-bootstrap 16–84% bands on the total; (b) the sparse B2/B4/B6/B8 sampling of the same profile under both channel-to-layer parity hypotheses, drawn as a nuisance envelope because the offset is unresolved (neither the even map in the hardware record nor the legacy odd map carried by the historical readout columns is canonical); (c) the proxy Sample I/II populations. Energy is raw deposited MeV, not Birks-visible.

6 Inter-stave timing

6.1 Observable and covariance

For staves i and j , define

$$\Delta t_{ij} = t_j - t_i - t_{\text{TOF},ij}.$$

The pair width does not uniquely determine single-stave resolutions. In general,

$$\text{Var}(\Delta t_{ij}) = \sigma_i^2 + \sigma_j^2 - 2 \text{Cov}(t_i, t_j).$$

Common trigger phase, sampling phase, reconstruction choices and event conditions may produce covariance. The paper therefore does not divide a pair width by $\sqrt{2}$ without an independently justified model.

6.2 Pulse-component definition

Leading-edge timing is amplitude-dependent. Constant-fraction timing also becomes ambiguous for multi-component waveforms if the threshold is defined from the global maximum: changing the fraction can switch the timed physical component. The production method must define the selected pulse component before optimising any timing fraction. Same-sample minimum widths across a fraction scan are exploratory model-selection diagnostics, not performance measurements.

6.3 Current 8-by-16 data diagnostic

On the located 8-by-16 raw waveform product, the current direct B4–B6 CFD reconstruction returns 5,207 events with valid times and an approximately 38 ns central-68% residual width (format-/reconstruction-limited; see the evidence box below) after the applied TOF subtraction. Peak positions are multi-modal and frequently lie near waveform boundaries. The component-safe rerun has been executed twice. The first rerun used the locked v1 polarity convention and yielded 8.748 ns on 10,776 events; a polarity measurement study subsequently falsified v1 for channels 2–7, invalidating that number. Under the measured polarity map v2, on the complete authorising

Sample II population (runs 58–63 and 65; 228,697 complete pairs above the 1000 ADC amplitude cut), the component-safe constant-fraction mode (20% fraction, first-local-peak component selection) with the 0.312 ns B4–B6 time-of-flight correction yields a pair-residual central-68% width of 0.146 ns (1,000-replicate bootstrap interval 0.144–0.148 ns) with a median of -9.67 ns, superseding both earlier diagnostics. Publication promotion remains conditional: the synthetic two-pulse component-recovery and wrong-component validations now pass under v2, but the constant -9.7 ns inter-channel offset is an uncalibrated skew, real-pulse multi-component identity is not validated on beam data, and timing/TOF definitions remain analysis contracts rather than source-bound hardware quantities.

Evidence boundary. The pair-residual width (0.146 ns under measured polarity v2, superseding the 8.748 ns v1-based rerun and the earlier 38 ns direct-mode diagnostic) is reported as a **pair residual at fixed amplitude selection**. The width lies far below the 10 ns sampling interval, which indicates strong sample-interpolation precision on high-SNR pulses above the 1000 ADC cut rather than a validated detector resolution; the constant -9.7 ns median offset, window phase, pulse-component ambiguity on real pulses, boundary truncation and timing-reference/TOF assumptions remain entangled. No intrinsic stave timing resolution is extracted, and the residual width is not divided by $\sqrt{2}$. The timing-residual calculation follows a manifest-tracked, version-controlled procedure; its authorising numerical record is bound as a gated figure below.

Gated figure – not rendered in the publication PDF

Component-safe rerun under measured polarity v2: central-68% width 0.146 ns (bootstrap 0.144–0.148 ns), 228,697 Sample II pairs, TOF-corrected; synthetic validations pass, but the -9.7 ns channel-skew offset and real-pulse component identity remain unresolved, so the claim stays gated and the figure is not rendered in the publication build.

Figure 6: B4–B6 pair timing residual from the source-bound 8-by-16 waveform product (Sample II, component-safe CFD20, first-local-peak mode). NOT DETECTOR RESOLUTION.

Historical sub-nanosecond values are retained only as provenance history or simulation/method-closure diagnostics and must not appear in the abstract/conclusions as beam-data performance.

7 Amplitude $\Delta E-E$ and the segmentation limit

7.1 Observable definition

The supervisor-defined beam-data analogue of a range-telescope observable is

$$\Delta E_{\text{data}} = A(B2), \quad E_{\text{data}} = A(B4) + A(B6) + A(B8),$$

where A is a baseline-subtracted ADC amplitude. These axes are response proxies, not calibrated energies.

A data-matched MC view should use the same four physical readout positions,

$$\Delta E_{\text{MC},4} = E_{\text{dep}}(B2), \quad E_{\text{MC},4} = E_{\text{dep}}(B4) + E_{\text{dep}}(B6) + E_{\text{dep}}(B8),$$

while a full-segmentation truth view must sum each unique downstream physical layer exactly once. A B2-versus-B4 plot is a two-channel response diagnostic and must not be called $\Delta E-E$.

7.2 Sparse-segmentation mechanism

Alternating readout positions can miss the layer in which the stopping-power rise becomes largest. A change in entry energy, angle or passive material can move the Bragg-rise region between sampled and unsampled layers and thereby alter the apparent sparse ΔE – E topology. The publication test is therefore a controlled comparison of full physical layers and the exact data-matched readout mask, with readout phase treated as a nuisance until the hardware map is source-bound.

7.3 Cycle-3 status of the current production bundle

PUBLICATION HOLD. The numerical bundle remains **not** publication-authorising. The producer was repaired: (i) beam-data input now uses the authorising 8-by-16 raw schema; (ii) threshold-censoring order is corrected; (iii) physical-layer and readout namespaces are kept distinct. Corrected-run evidence is recorded per claim in the ledger. Gate status as of 2026-08-14: (iv) the channel-to-layer map remains a documented simulation contract, not a hardware-source-bound drawing; (v) the saturation and transfer threshold remains a software contract without hardware source-binding (recorded as a preserved blocker; no hardware identity is invented); (vi) MC production provenance is closed: the regenerated 100,000-event campaign carries a complete production receipt in a machine-readable manifest. The quarantine on the OLD bundle stays; no promotion to validated status is authorized until all gates pass and a final figure package is delivered.

The quarantined plots are kept in a separate gated figure area to preserve provenance without allowing accidental use in the paper.

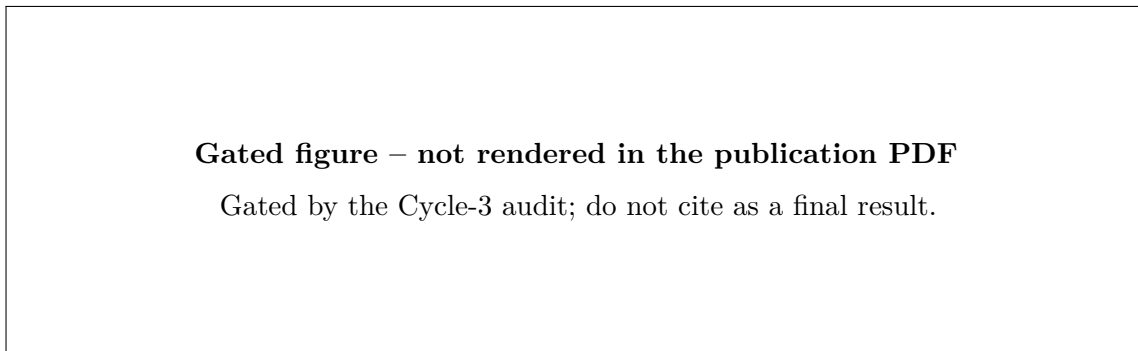


Figure 7: Quarantined current beam-data amplitude ΔE – E plot.

The final Figure 7/8 production must start from an event-level table that contains all four reconstructed channel amplitudes before analysis-threshold censoring, then apply declared threshold scans; import the source-bound readout map; keep physical-layer and readout namespaces distinct; and use a production MC whose source measure and run provenance are authorising.

7.4 Species-resolved topology and penetration deliverables

The supervisor-requested species and penetration set is delivered from the corrected event tables of the regenerated-campaign producer. Species-resolved ΔE – E panels (proton truth, deuteron truth, colour-coded proton+deuteron, and all particles) are produced for Sample I and Sample II in both the full downstream-sum and the data-matched four-layer definitions. Penetration-depth distributions (L_{stop} : deepest physical layer with a deposit above a stated threshold, with a threshold-stability scan) are produced per species and per sample, together with the fraction of events reaching each downstream readout position (B4/B6/B8) and the beam-data Sample I versus Sample II stopping-layer overlay from the producer’s threshold-pass logic. Beam-data axes remain amplitude

proxies (not calibrated energy); B2 saturation is shown only as the analysis-contract flag, with no numeric hardware level asserted.

The MC panels are Geant4 deposited-energy truth views from a 2,000,000-event scale-up of the regenerated campaign (recorded in a machine-readable manifest; identical sampler, geometry and runtime stack as the authorising 100,000-event campaign). Because both campaigns run the default engine seed over the same sequential stream, the scale-up replicates the authorising campaign as a verified sequential prefix: the leading B-stack-entrance event rows are identical event-by-event, so the added 1.9 million events extend the same random stream (the leading 100,000 are duplicated, not independently redrawn) and only densify the same truth topology. The set is descriptive truth topology, not a detector-response prediction: readout thresholds, light collection and digitisation are not in the loop, so no numeric agreement with the amplitude-domain data figures is claimed or implied. Beam-data figures remain gated under the same Cycle-3 gates as Figure 7/8; the MC truth views are descriptive simulations and promote no quarantined number.

MC truth ΔE - E Sample I (full downstream sum), unweighted event counts

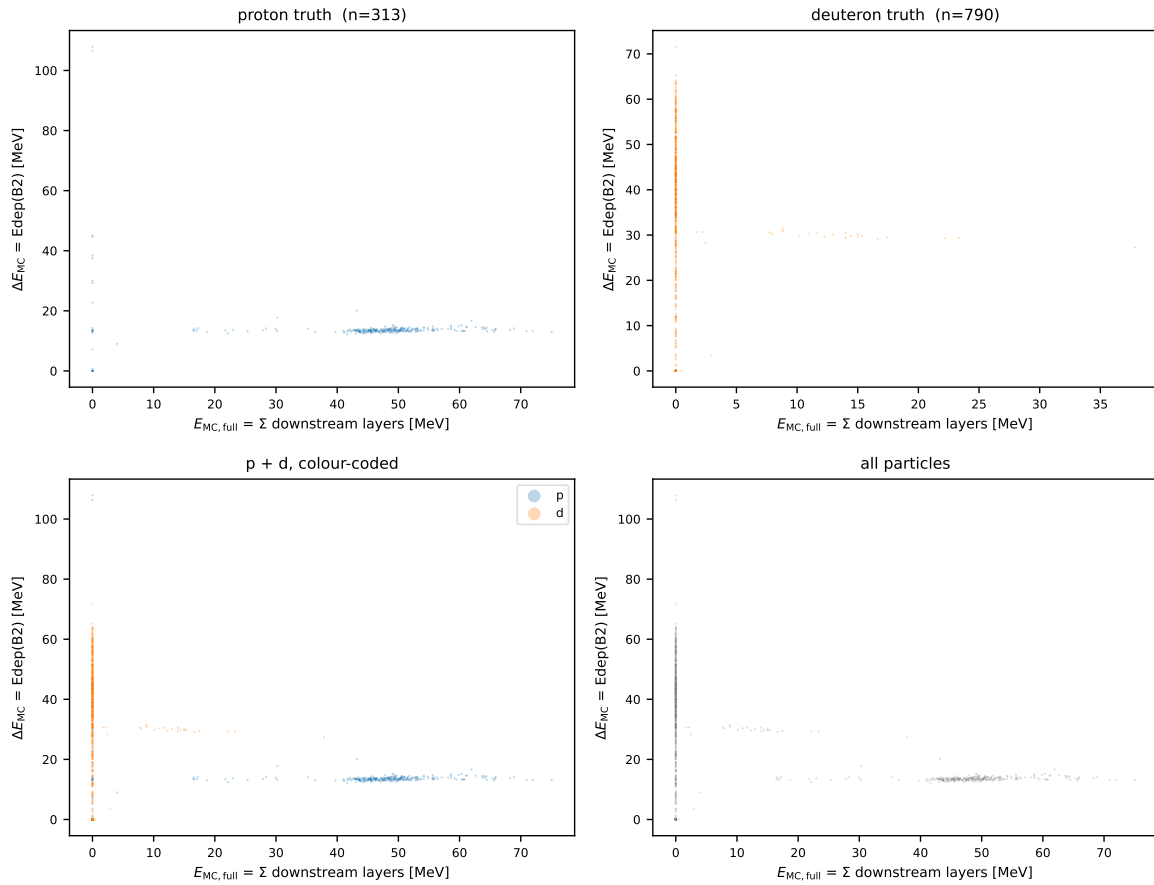
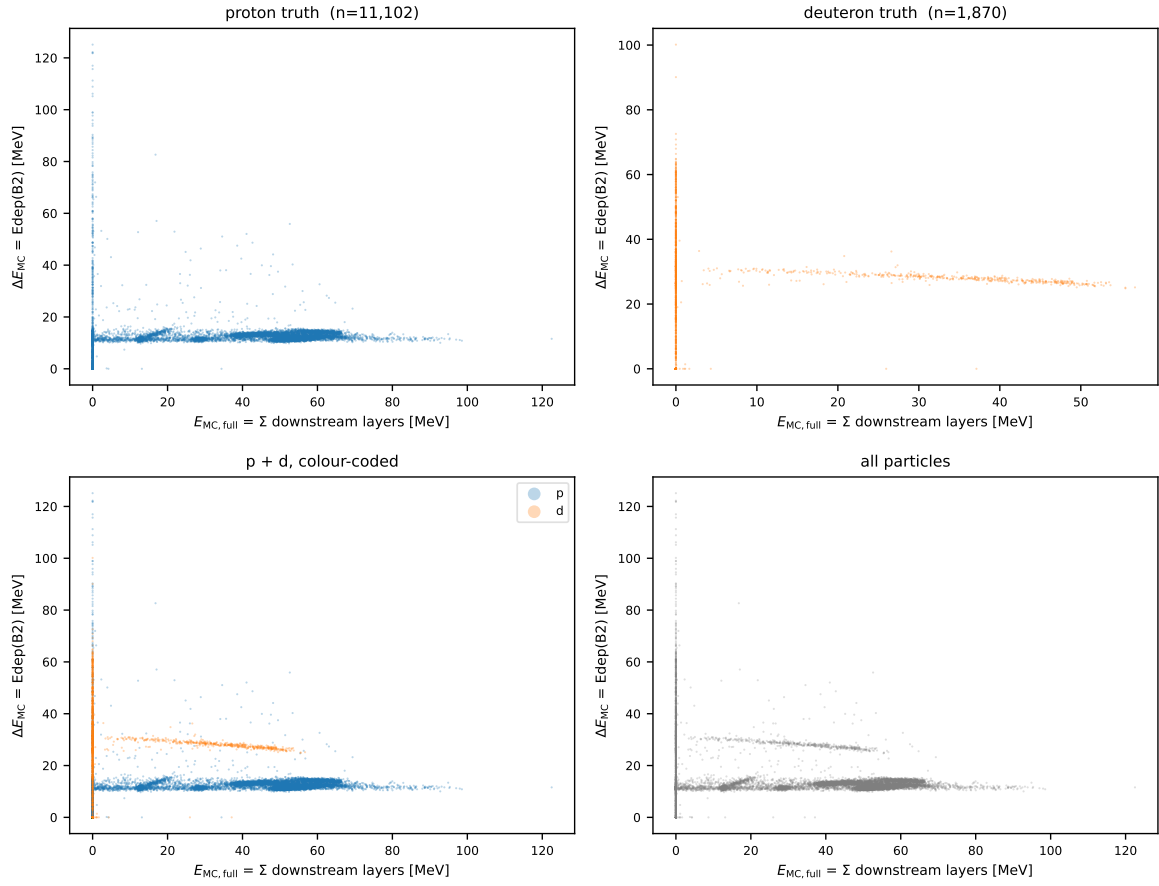
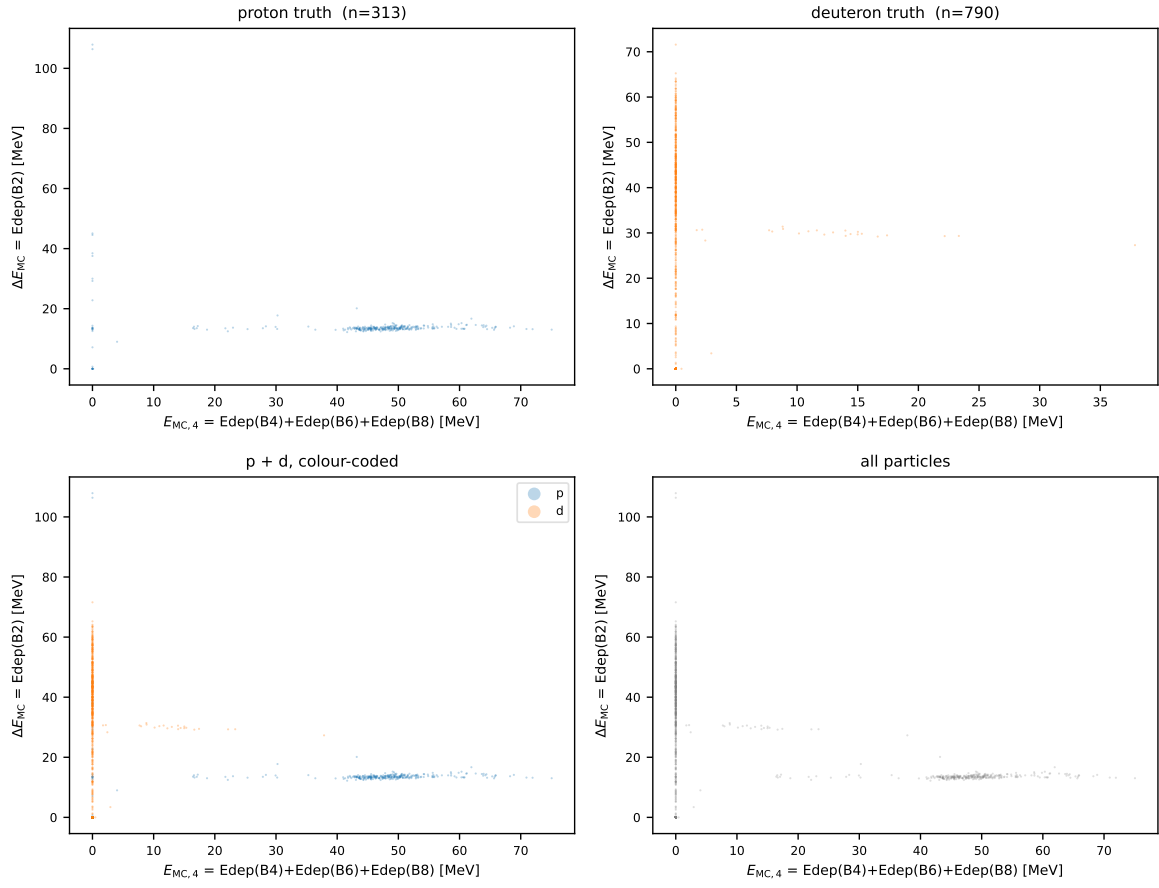
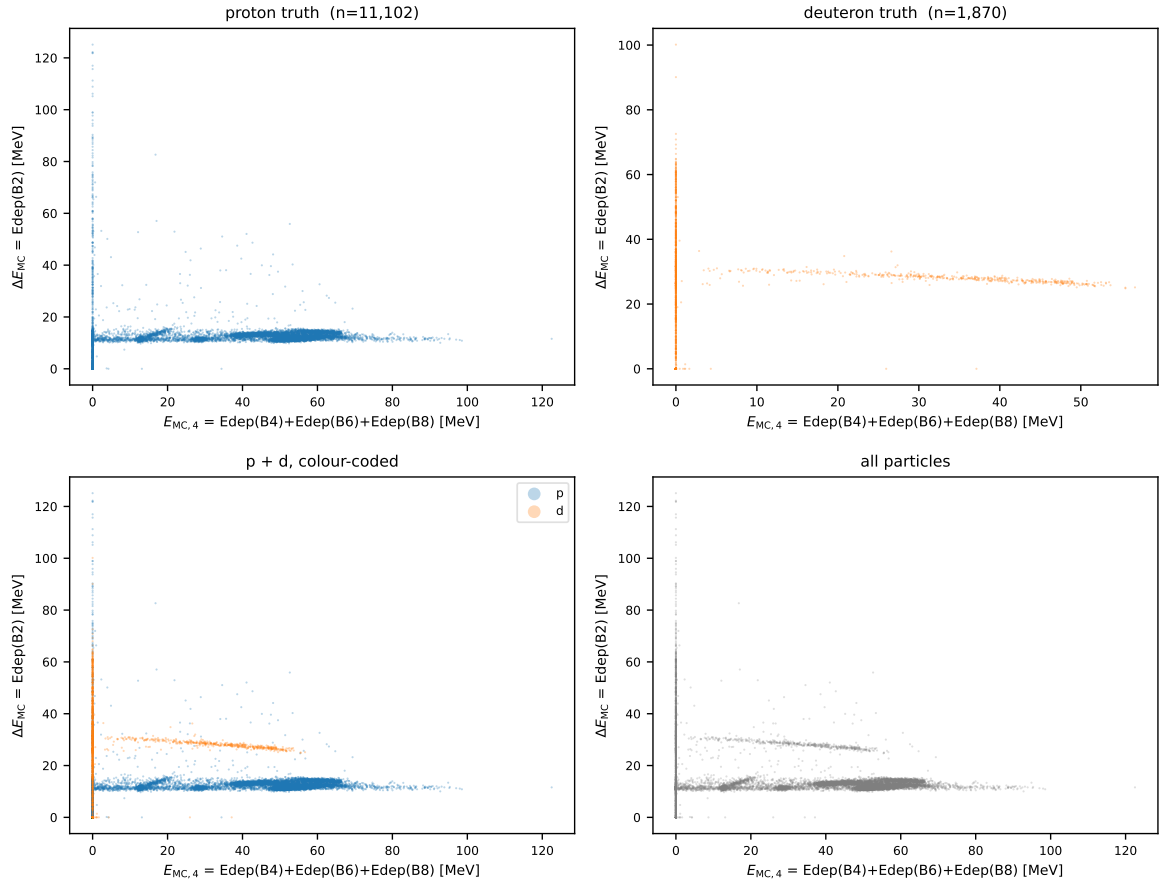


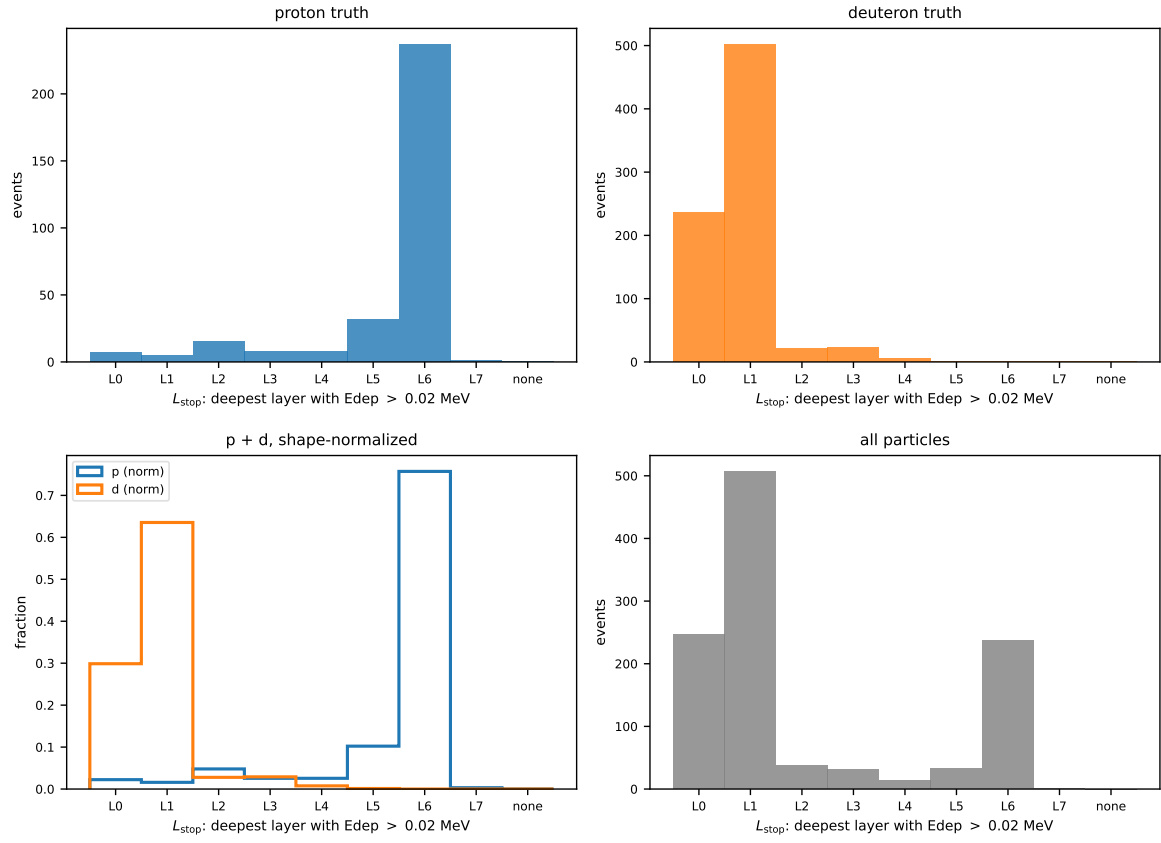
Figure 8: MC truth ΔE - E , Sample I, full downstream sum; proton/deuteron/combined/all panels (descriptive simulation).

MC truth ΔE - E Sample II (full downstream sum), unweighted event countsFigure 9: MC truth ΔE - E , Sample II, full downstream sum; same panel structure (descriptive simulation).

MC truth ΔE -E Sample I (data-matched 4-layer), unweighted event countsFigure 10: MC truth ΔE -E, Sample I, data-matched four-layer definition (descriptive simulation).

MC truth ΔE - E Sample II (data-matched 4-layer), unweighted event countsFigure 11: MC truth ΔE - E , Sample II, data-matched four-layer definition (descriptive simulation).

MC penetration depth, Sample I (unweighted)

Figure 12: MC penetration depth L_{stop} , Sample I: proton, deuteron, shape-normalised overlay, all particles (descriptive simulation).

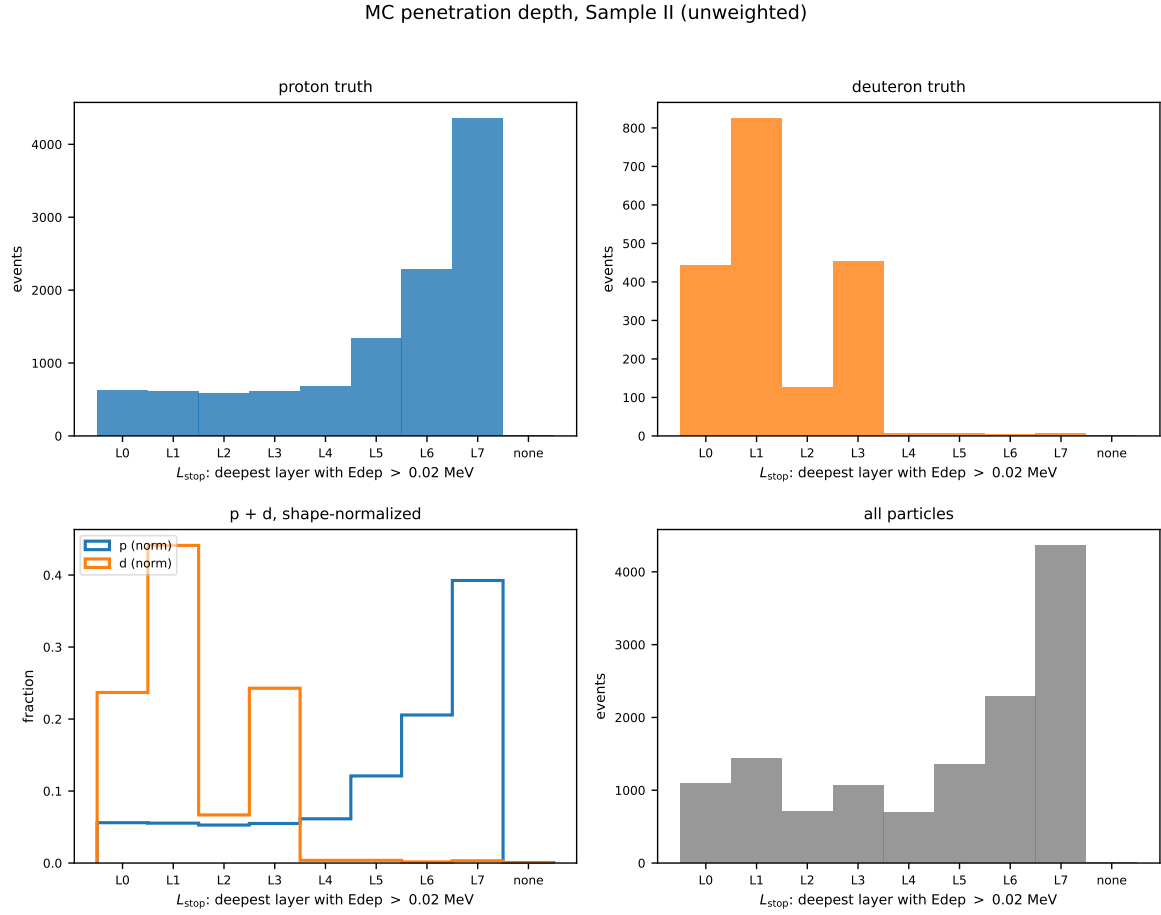


Figure 13: MC penetration depth L_{stop} , Sample II, same panel structure (descriptive simulation).

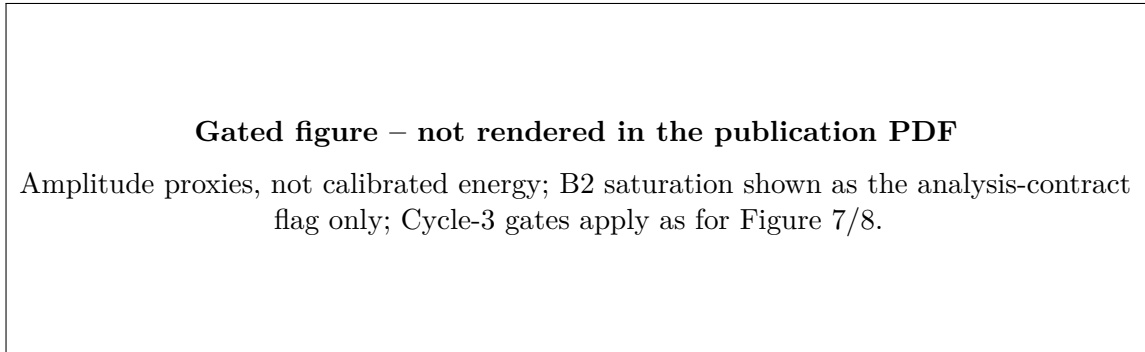


Figure 14: Beam-data amplitude analogue $\Delta E - E$, Samples I and II on identical axes (gated).

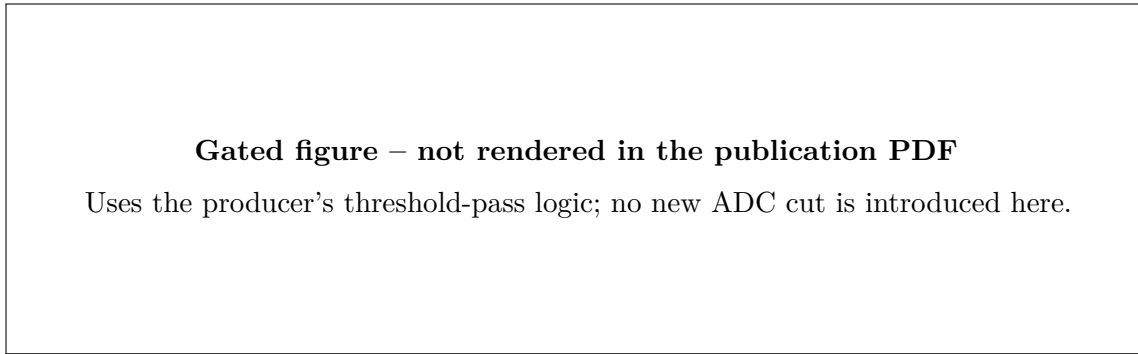


Figure 15: Beam-data stopping-layer overlay, Sample I versus II, amplitude domain (gated).

8 Monte Carlo prediction of light collection and transport

8.1 Stage-resolved response

The optical model should report separately the production of scintillation photons, WLS absorption, WLS re-emission, guided/transported photons, sensor arrivals, primary SiPM avalanches, correlated avalanches, charge and ADC response. Useful conditional efficiencies include

$$\epsilon_{\text{WLS capture}} = \frac{N_{\gamma, \text{WLS abs}}}{N_{\gamma, \text{scint}}}, \quad \epsilon_{\text{transport}} = \frac{N_{\gamma, \text{sensor}}}{N_{\gamma, \text{WLS emit}}}, \quad \epsilon_{\text{PDE}} = \frac{N_{\text{primary aval}}}{N_{\gamma, \text{sensor}}}.$$

These ratios should be studied versus hit position, species/deposit regime and model nuisance parameters. Their denominators must be named explicitly: an efficiency conditional on WLS emission is not an absolute scintillator-to-sensor efficiency.

8.2 Historical July calibration grid

The July single-stave grid contains five proton/deuteron points and historically produced an order-ten detected-PE per visible-MeV scale. It is useful as a regression/history object, but it is not the nominal publication model.

PUBLICATION HOLD. The exact July calibration revision predates multiple source-level fixes. In particular, inner and outer fibre claddings both used the shared NIST G4_PLEXIGLASS material and attached different refractive-index tables, so that the later assignment overwrote the intended inner-clad optical properties. The campaign also predates explicit WLS fluorescence-yield/multiplicity, separated scintillator/Y-11 attenuation controls, corrected reflector-surface semantics, source-bound fibre-to-SiPM interface modelling and a single canonical SiPM response state. Its historical “PE/MeV” denominator is the Birks-visible variable, not unquenched deposited energy. The grid therefore required regeneration before any numerical light-yield result is promoted.

8.3 Regenerated 2026-08 calibration grid (current model)

The five-point grid was regenerated (p60/p100/p140/d70/d110, 2000 events per point, seed-bound, distinct inner/outer cladding materials, SHA-256-bound inputs and manifests). The pooled Birks-visible light yield is 11.78 PE/MeV_{vis} (offset 0.65 PE, $r^2 = 0.976$, $n = 10,000$); per-point E_{vis} yields are flat within their bootstrap intervals (11.78–11.84), while E_{raw} yields (8.6–10.6) split by species through the Birks quenching ratio ($E_{\text{vis}}/E_{\text{raw}} \approx 0.81$ for 110 MeV deuterons). Stage accounting

gives mean capture $\epsilon_{\text{WLS}} \approx 0.19$, transport ≈ 0.021 , detection-given-arrival ≈ 0.30 , and a saturated-PE-to-detected ratio ≈ 0.96 at the readout channel. These numbers are MC-model-dependent: the optical/SiPM nuisance envelope is not yet propagated, so the July grid remains the historical object and no beam-data calibration is claimed. Machine-readable source tables accompany every figure.

A subsequent audit of the grid build revisions (both descendants of the 2026-08-11 change that introduced the `WLSMEANNUMBERPHOTONS` property) shows the 2026-08 grid actually sampled *Poisson*(1) WLS re-emission multiplicity per absorption while its run metadata labelled the model `geant4_default_one_secondary`: in Geant4 11.2.2 the WLS process samples Poisson whenever that property exists, even at mean 1. The pooled mean light yield is unaffected in expectation ($E[\text{Poisson}(1)] = 1$), but per-event multiplicity variance and saturation-adjacent observables carry Poisson(1) semantics under a default-one label. A later correction implements an explicit three-mode multiplicity contract (exactly-one, Poisson(μ), Bernoulli(q) with $q = 0.70$ from the Y-11/K27 quantum yield; Pla-Dalmau, Foster, Zhang, NIM A361 (1995) 192–196), makes the optical digest mode-complete, and adds a known-answer test that discriminates the modes; grid regeneration under the corrected contract is tracked as a follow-up.

The regenerated grid evidence objects themselves are truth-level MC records and are rendered here with their model-dependence boundary stated on-caption. They are simulation results of the current optical model, not a calibrated light-yield claim.

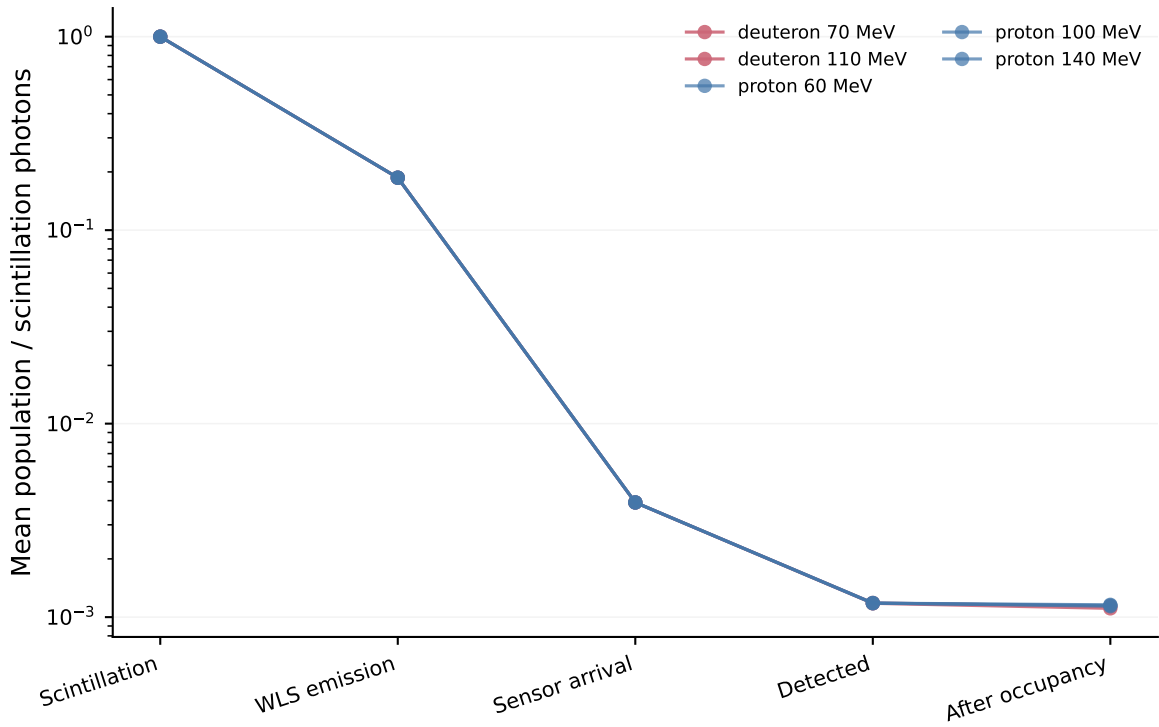


Figure 16: Regenerated five-point optical grid: stage-resolved accounting from scintillation production through WLS absorption/re-emission, guided transport, sensor arrival and detection to saturated-PE readout. Mean stage ratios: WLS capture ≈ 0.19 , transport ≈ 0.021 , detection-given-arrival ≈ 0.30 , saturated-to-detected ≈ 0.96 .

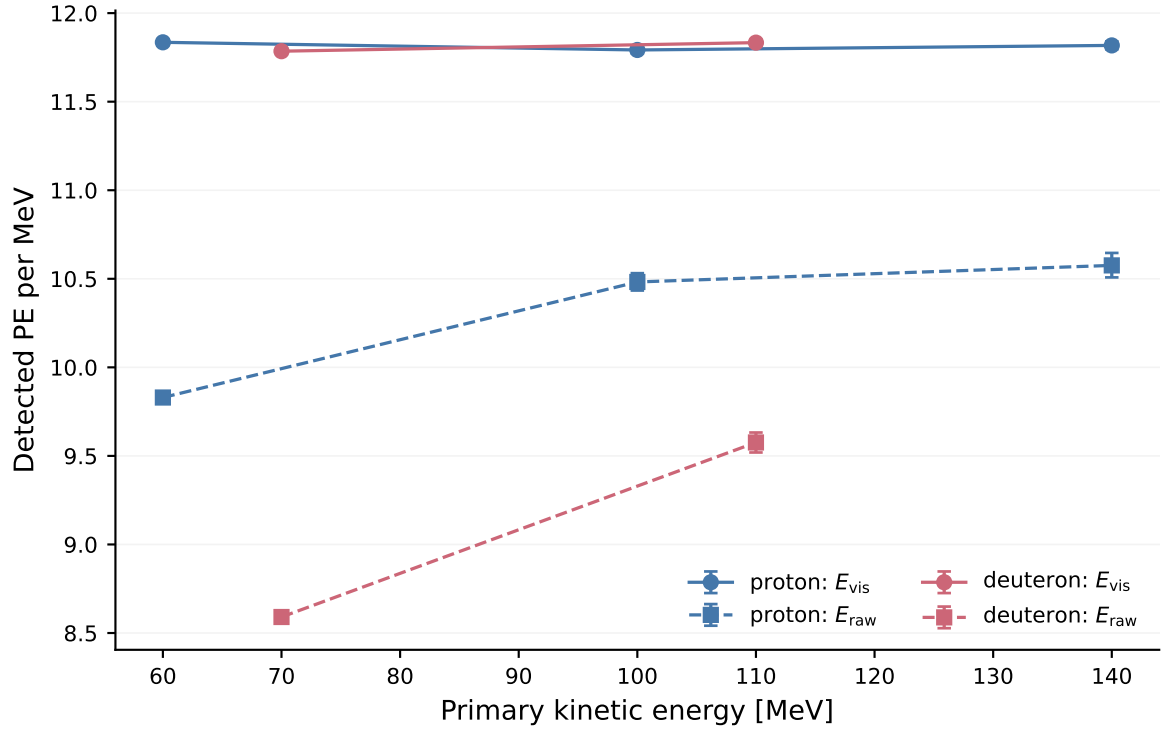


Figure 17: Per-point detected-PE yields per visible MeV (flat within bootstrap intervals, 11.78–11.84) and per raw deposited MeV (8.6–10.6, species-split through the Birks quenching ratio) across the five-point grid.

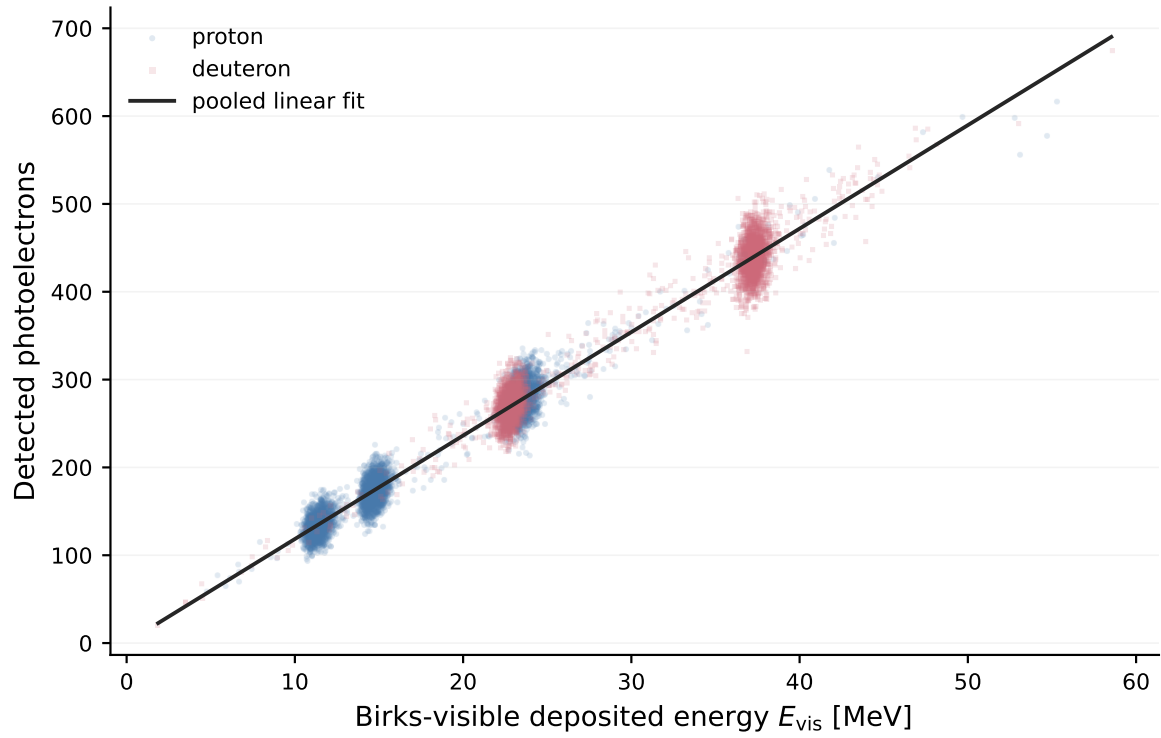


Figure 18: Detected PE versus E_{vis} and E_{raw} with pooled linear fits (vis: 11.78 PE/MeV_{vis}, offset 0.65 PE, $r^2 = 0.976$, $n = 10,000$).

The historical values and plots remain available in the gated figure area and the source report for

correction history only.

8.4 Required nominal optical campaign

A publication campaign must bind a clean source revision and build receipt; preserve E_{raw} and E_{vis} separately; record stage counters; use the declared one-fibre/one-end design channel as the nominal response only after the installed hardware configuration is source-bound; retain alternative fibre/end observables as model controls where useful; scan or state the support in longitudinal/transverse position and angle; and propagate independent high-impact optical nuisance families rather than re-tuning the total mean response back to a historical target.

9 Energy reconstruction and resolution

9.1 Measurand definition

Energy resolution is not a single detector-independent number. For the single-stave response chain, two physically distinct targets are available:

$$E_{\text{raw}} \equiv \text{unquenched Geant4 energy deposited in the scintillator}, \quad (1)$$

$$E_{\text{vis}} \equiv \text{Birks-visible/quenched energy used to generate scintillation light}. \quad (2)$$

A full range-stack reconstruction may later target incident charged-particle kinetic energy from amplitudes and stopping depth. These estimands must not share one bare E_{dep} label.

For a chosen target E_T , define

$$r = \frac{E_{\text{reco}} - E_T}{E_T}.$$

A publication result should report median bias, σ_{68} , RMS, tail fraction and uncertainty intervals on held-out populations, as functions of target energy and relevant phase-space variables. The uncertainty unit must reflect the independent simulation/run conditions used to establish transfer, not merely the number of events generated under a few shared grid settings.

9.2 Status of the current held-out result

PUBLICATION HOLD. The 8.9% held-out σ_{68} result remains quarantined. Energy semantics are now enforced throughout the analysis stack: E_{raw} (unquenched Geant4 scintillator deposit) and E_{vis} (Birks-visible quenched energy) are distinct estimands. The held-out producer was rewritten with proper estimand definition and propagated uncertainty, and the underlying optical grid was regenerated (2026-08). On the regenerated five-point grid with the E_{vis} target: held-out median bias +0.48% [+0.29, +0.62], $\sigma_{68} = 6.26\%$ [6.16, 6.35], tail fraction 0.2%; the E_{raw} estimand (deliberately mis-specified pooled-linear response) gives bias +10.4%, $\sigma_{68} = 9.0\%$, tail 16.9% and is the negative control, alongside the shuffled-target control reported in the result manifest. The control's mis-specification is species stratification in the E_{raw} domain: the per-species detected-PE versus E_{raw} responses are intercept-dominated and species-split (deuteron 4.62 PE/MeV_{raw} + 198.5 PE, proton 5.11 PE/MeV_{raw} + 80.9 PE over the ~17–30 MeV deposit range), so no single pooled E_{raw} line represents both (pooled: 7.47 PE/MeV_{raw} + 49.0 PE; a species-aware E_{raw} refit degrades to −9.3% median bias with $\sigma_{68} = 47.3\%$). The E_{vis} estimand is species-blind by construction (per-species slopes 11.79 and 11.65 versus pooled 11.77 PE/MeV_{vis}; species-aware metrics unchanged at +0.7% and 6.3%). Bootstrap 16–84% intervals use 500 resamples of the held-out population. The result remains MC-model-dependent and quarantined from promotion to validated status: the optical/SiPM nuisance envelope is not yet propagated, held-out coverage spans two operating points only, and impact-position transfer is scoped to the central line ($y = 0$). No promotion to validated status is authorized.

Gated figure – not rendered in the publication PDF

Gated: wrong/ambiguous energy semantics, incomplete uncertainty and superseded optical campaign.

Figure 19: Quarantined Figure 11 from the historical held-out reconstruction study.

Gated figure – not rendered in the publication PDF

Gated: MC-model-dependent (optical/SiPM nuisance envelope not propagated; two held-out operating points; simulation, not beam data).

Figure 20: Held-out energy reconstruction on the regenerated optical grid (E_{vis} estimand): residual distribution for held-out deuteron_110 and proton_60 with bootstrap 16–84% intervals; median bias +0.48%, $\sigma_{68} = 6.26\%$, tail fraction 0.2%.

Gated figure – not rendered in the publication PDF

Gated: deliberately mis-specified response control; shown to demonstrate estimand separation, not as a performance claim.

Figure 21: Negative control: reconstruction against the E_{raw} (unquenched) target using a deliberately mis-specified pooled linear response fitted in the E_{raw} domain (median bias +10.4%, $\sigma_{68} = 9.0\%$, tail fraction 16.9%).

9.3 Required publication analysis

The replacement study should preregister and freeze training/calibration, model-selection and final-evaluation populations before fitting; include multiple well-separated target-energy regimes; validate particle-species transfer without truth-label tuning; test longitudinal/transverse position and angle support; compare transparent linear/monotonic baselines before any ML model; and propagate optical/SiPM model nuisance families separately from finite-simulation statistics. Bootstrap or hierarchical intervals should resample the independent grid/run unit appropriate to the claim. A shuffled-target or deliberately mis-specified response control should fail the reconstruction test,

and occupancy/saturation diagnostics should be checked on known-answer inputs with a signed definition. If a resolution parametrisation is fitted, the available independent energy grid must identify its terms.

A full-stack incident-energy estimator may use the amplitude vector $\mathbf{A} = (A_{B2}, A_{B4}, A_{B6}, A_{B8})$ and a source-bound penetration/range observable, but it remains downstream of material-budget, trigger/selection and per-channel response closure.

10 Discussion

Existing selected-data studies indicate that the declared Sample-I and Sample-II populations occupy the B stack differently in depth. The difference has now been carried through the pre-threshold 8-by-16 event reconstruction with run-block uncertainty and a baseline/threshold scan. This is the paper's most direct experimental range-stack observation: its direction is stable across the 0–1000 ADC scan while its magnitude is strongly threshold-compressed. Even so, the first interpretation remains a run-family dependence: a unique trigger-causation statement additionally requires the primary hardware trigger and run log to be source-bound.

Sparse longitudinal sampling is a central detector-design issue. An alternating four-position readout can preserve a coarse penetration pattern while losing the local sequence of energy loss near the Bragg rise. This creates a concrete aliasing mechanism: changes in particle energy, incidence angle or passive material can move the high- dE/dx region between instrumented positions. A quantitative PID or range estimator based on sparse ΔE – E information therefore requires either sufficiently dense longitudinal instrumentation or an independently validated model of the missing-layer response, with the readout phase treated as a nuisance rather than as hidden truth.

The timing analysis gives a complementary negative result. In the component-safe rerun under the measured polarity map, the B4–B6 pair residual has a central-68% width of 0.146 ns (bootstrap interval 0.144–0.148 ns) with a constant -9.7 ns channel-skew median, superseding both the earlier 38 ns direct-mode diagnostic and the 8.748 ns value produced under the falsified polarity convention. This width is a high-SNR sample-interpolation precision at fixed amplitude selection, not an intrinsic stave resolution, and it is not divided by $\sqrt{2}$. Component identity, timing reference, run dependence and channel covariance must be established before any deconvolution to a single-stave resolution is attempted. This limitation is scientifically useful because it states what a future campaign must record or calibrate: a source-bound timebase, sufficient waveform window, stable component assignment and an independent timing reference or covariance model.

The optical-model audit shows why a detector-response study must distinguish a causal response graph from a tuned total efficiency. WLS fluorescence multiplicity, trapping and attenuation, fibre-end coupling, SiPM photon-detection response, microcell occupancy/recovery, correlated noise and electronics can compensate one another in an integrated charge or amplitude. Stage-resolved counters, phase-space scans and independent bench/hardware constraints are therefore more informative than tuning a single PE/ADC conversion to one dataset.

The present limitations are coupled rather than independent. Geometry and run/trigger uncertainty affect which particles enter the selected data population; source-model and event-measure uncertainty affect the MC comparison; sparse readout affects the observability of a local stopping-power rise; and optical/electronics assumptions affect any conversion from deposited energy to charge or ADC. The final paper should therefore report the direct data observables first and introduce model-dependent detector-performance claims only after these nuisance layers have been propagated through the same final population definitions.

11 Conclusions

The CCB test-beam analysis provides a controlled test of a segmented plastic-scintillator range stack for low-energy charged particles. The regenerated pre-threshold 8-by-16 event population (33 runs, 1.1M events, run-block bootstrap) confirms different longitudinal B-stack populations for the declared Sample-I and Sample-II run families (entrance share 35.3% versus 15.3% in B2 and a B8/B2 amplitude-share ratio rising from 0.71 to 1.94, with the direction preserved under both duplicate-readout hypotheses), but the quantitative depth-profile claim remains gated: the profile shape is strongly threshold-dependent and the primary hardware-trigger record is not source-bound. The analysed B readout samples only alternating longitudinal positions, so any range or energy-loss estimator is sensitive to where the stopping-power rise falls relative to instrumented positions. The intended telescope-style amplitude observable is $\Delta E = A(B2)$ and $E = A(B4) + A(B6) + A(B8)$; a B2–B4 plot is a two-channel diagnostic rather than the complete downstream-energy proxy.

The located 8-by-16 waveform product supports a negative timing conclusion rather than a detector-resolution number. In the component-safe rerun under the measured polarity map, the B4–B6 pair residual has a central width of 0.15 ns with a -9.7 ns channel-skew offset (superseding an 8.7 ns value produced under the falsified polarity convention and an earlier 38 ns direct-mode diagnostic); the width is not converted to a single-stave resolution because component identity, timing reference and covariance terms remain unresolved. Historical sub-nanosecond simulation or incompatible-waveform results must therefore not be transferred to this beam-data product.

A single-stave Geant4 optical model remains valuable as a response framework. The historical July calibration campaign is superseded as the nominal model; the grid has been regenerated on the repaired source with explicit E_{raw} and E_{vis} semantics, and the held-out reconstruction has been rebuilt on it with propagated uncertainty. These regenerated results remain MC-model-dependent and quarantined from validated promotion: the optical/SiPM nuisance envelope is not yet propagated, held-out coverage spans two operating points only, and impact-position transfer is scoped to the central line, so they do not authorize absolute light-yield or resolution claims.

The paper is therefore not yet a quantitative detector-performance publication. The shortest defensible path to submission is to close the source-bound hardware/run/polarity and 8-by-16 data contracts first, produce the final longitudinal beam-data result, establish a provenance-complete MC event measure, close the remaining ΔE – E gates (hardware source-binding of the channel-to-layer map and of the saturation/transfer threshold) and deliver the final figure package, and then decide whether the optical/energy programme is mature enough to remain in the same paper. If it is retained, the remaining optical steps are propagation of response-stage nuisance families and a broader genuinely held-out reconstruction study across more operating points and impact positions; the regenerated grid already separates E_{raw} and E_{vis} . Every headline result should then be bound to its machine-readable source table, uncertainty, canonical claim status and exact reviewed source head before submission.

12 Reproducibility and figure-generation requirements

Every quantitative paper figure must be generated from a tracked machine-readable result or source table. The producing record must include source commit, producer hash, input paths and SHA-256 values, event/run population, selection, weights, uncertainty method and evidence class. No headline number should be manually copied into a plotting script or caption.

Data and MC must use compatible observable definitions. Beam ADC amplitudes and truth-level Geant4 energy deposits may be shown in separate panels, but they must not be placed on a shared calibrated energy axis without a validated response transformation. Weighted MC results must state the weight semantics and report $\sum w$, $\sum w^2$, effective sample size and weight pathologies relevant

to the estimand.

Timing and energy calibrations must be fit on declared calibration/training populations and evaluated on held-out runs or simulation populations without retuning. Resampling must preserve the physical/statistical source unit, normally DAQ run/event for beam data and generator event for MC, rather than assuming every exported hit/pulse row is independent.

The canonical publication workflow is

physics/provenance contract $\rightarrow$ result + manifest + source table
 $\rightarrow$ claim ledger $\rightarrow$ figure/table
 $\rightarrow$ manuscript $\rightarrow$ adversarial review.

A result is not publication-complete when code alone runs successfully.

A Publication status and open gates

This appendix records the scientific readiness state of the working manuscript. It is collaboration-review metadata rather than a substitute for the final scientific results narrative; before journal submission, detailed gate bookkeeping moves to supplementary reproducibility material.

Gate	Publication consequence
Installed-hardware record	Source-bind installed stave, channel map, target, trigger and survey/run settings.
Run roles & trigger response	Prevents over-interpreting Sample-I/Sample-II differences as pure trigger effects.
Readout map & waveform polarity	Required before authorising event-level amplitude observables.
Beam/MC $\Delta E-E$ regeneration	Regenerate beam/MC $\Delta E-E$ from the authorising pre-threshold 8×16 event product and corrected MC layer namespaces.
MC measure, source uncertainty & provenance	The deterministic direct-CDF sampler defect is repaired and closed; that bounded software closure does not authorise the historical paper MC.
WLS photophysics & optical grid	Regeneration of the single-stave optical grid after superseded source defects, followed by replacement optical/energy figures.
Energy semantics & held-out study	Raw deposited energy vs Birks-visible energy semantics and a corrected held-out reconstruction study.
Figure packages & manifests	Setup, longitudinal-profile and timing figures are wired into the submission manifests with SHA-bound machine-readable sources; the beam-data profile and timing artifacts remain gated.
Single source of truth	One scientific claim-status record across figures and manuscript, plus stale-document reconciliation.
Submission fail-closed gates	Umbrella submission and merged-state fail-closed checks.

As of the 2026-08-17 manifest wiring, the final-figure namespace carries 15 figures and the final-table namespace 4 tables, each with artifact SHA-256 and a machine-readable source table; the source-data namespace carries the small per-figure source tables. A mechanical validator enforces this contract. The remaining violations are the known scientific blockers: the gated claims behind the beam-data profile, timing residual and held-out reconstruction summaries, the standing publication-hold record, and the build receipt predating the current head.

The Cycle-3 audit documents the detailed failure modes in its audit record and addenda; the current-head dependency order and the section-by-section review are recorded in the corresponding review records.

B Evidence surfaces and provenance

The working paper is derived from, and must remain synchronized with, the following categories of collaboration evidence:

- a canonical scientific claim-status record;
- the publication hardware bill of materials (truth/provenance surface);
- the figure registry, currently subject to governance repair;
- the fail-closed submission review record and its addenda;
- the located raw beam-data audit and its timing limitation;
- the source-level CCB Geant4 review;
- the absolute WLS-yield boundary record;
- the SiPM-response boundary record.

A previous Markdown working manuscript is retained for provenance; the chapterised TeX files are the canonical editing surface.

References

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